

PDFs & CDFs

CSU, Chico Math 350

2026-04-15

outline

Probability Density Functions (PDF)

Cumulative Distribution Function (CDF)

References

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Concept

Random variables that can assume every value in an interval have continuous distributions. A continuous distribution can also be characterized by its probability density function (PDF).

Many experiments or observations of random phenomenon do not have integers as outcomes, but instead are measurements selected from an interval of numbers. For example, you could find the length of time that it takes when waiting in line at the grocery store or the weight of a bag of potato chips advertised at 1 oz.

If the measurements could come from an interval of possible outcomes, we call them **continuous** data.

Definition

A probability density function (PDF) is a function f such that:

- ▶ $f(x) \geq 0$ for all x
- ▶ $\int f(x)dx = 1$ over the domain of support

Flame-broiled example

Example 1

Suppose $f_Y(y) = 4y^3$, $0 \leq y \leq 1$. Is this a valid probability distribution?

Example 1

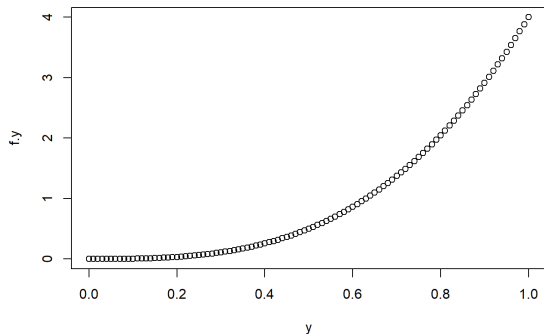
Yes, but you'll need to show this:

First: $f(y) \geq 0$ over the domain of support:

```
y <- seq(0,1, by=.01)
```

```
f.y <- 4*y^3
```

```
plot(y, f.y)
```



Example 1

Second: $f(y)$ integrates to 1 over the domain of support:

```
f.y <- function(y){4*y^3}  
integrate(f.y, lower=0, upper=1)  
## 1 with absolute error < 1.1e-14
```

The `integrate()` function is very useful for these operations.

HOWEVER, be certain that you are also able to integrate these by hand and show the work involved that the function is doing here!

Definition 4.3: Continuous Random Variable

A *continuous* random variable X is a random variable described by a pdf in the sense that:

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx$$

whenever $a \leq b$, including the cases $a = -\infty$ or $b = \infty$

Example 2

Let X be a random variable with the following PDF:

$$f(x) = \begin{cases} \frac{2}{3}x^{-1/3} & \text{for } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Compute $P(X \leq \frac{8}{27})$

Example 2

By hand:

$$\int_0^{8/27} \frac{2}{3} x^{-1/3} dx = x^{2/3} \Big|_0^{8/27} = \frac{4}{9}$$

Using R functions:

```
integrand <- function(y) { 2/3*y^(-1/3)}  
integrate(integrand,lower= 0 ,upper= 8/27)  
## 0.4444444 with absolute error < 6.7e-16
```

Exercise 1

a. Suppose $f_Y(y) = 4y^3$, $0 \leq y \leq 1$. Find $P(0 \leq Y \leq \frac{1}{2})$. Solve both by hand and using R.

b. For the random variable Y with pdf $f(y) = \frac{2}{3} + \frac{2}{3}y$ for $0 \leq y \leq 1$, find $P(\frac{3}{4} \leq Y \leq 1)$. Solve both by hand and using R.

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Concept

The cumulative distribution function (cdf) associated with X (either discrete or continuous) is the function $F(x) = P(X \leq x)$, or written out in terms of the PDFs and CDFs:

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t)dt$$

for continuous variables and

$$F(x) = P(X \leq x) = \sum_{n=-\infty}^x p(n)$$

for discrete variables.

Theorem 4.1

Let X be a continuous random variable with PDF f and CDF F .

- ▶ $\frac{d}{dx}F = f$
- ▶ $P(a \leq X \leq b) = F(b) - F(a)$
- ▶ $P(X \geq a) = 1 - F(a) = \int_a^\infty f(x)dx$

Additionally: $P(X = x) = 0$ no matter what, so
 $P(X \leq x) = P(X < x)$.

Example 3

A random variable Y has the following CDF:

$$F(y) = \begin{cases} 0 & \text{for } y < 1 \\ \ln(y) & 1 \leq y \leq e \\ 1 & y > e \end{cases}$$

- a. Find $P(Y < 2)$
 - b. Find $P(2 < Y \leq 2.5)$
 - c. Find $P(2 < Y < 2.5)$
 - d. Find $f(y)$
-

A)

By hand:

$\ln(2)$

Using R (to get exact value):

`log(2)` # note that `log()` means NATURAL LOG in R!

`## [1] 0.6931472`

Example 3

B)

By hand:

$$\ln(2.5) - \ln(2)$$

Using R (to get exact value):

$$\log(2.5) - \log(2)$$

```
## [1] 0.2231436
```

C)

Identical to B! Recall, $P(Y = 2.5) = 0$

D)

$$f(y) = \frac{d}{dy} F(y) = \frac{1}{y}$$

To clarify, recall that the derivative of any constant is 0, so the only non-zero derivative occurs when $F(y) = \ln(y)$, which has a derivative of $\frac{1}{y}$.

Exercise 2

a. A random variable Y has the following CDF:

$$F(y) = \begin{cases} 0 & \text{for } y < 0 \\ 4y^3 - 3y^4 & 0 \leq y \leq 1 \\ 1 & y > 1 \end{cases}$$

Find $P(\frac{1}{4} < Y < \frac{3}{4})$.

b. Suppose $F(y) = \frac{1}{12}(y^3 + y^2)$, for $0 \leq y \leq 2$. Find $f(y)$

c. Say the distribution of a family's disposable income, Y , is described by the PDF $f(y) = ye^{-y}$, $y \geq 0$. Find $F(y)$.

HINT: Integration by parts is useful here:

$$\int u * dv = u * v - \int v * du$$

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references I

Speegle & Clair. Probability, Statistics, and Data: A Fresh Approach Using R, 2021.

Also adapted from notes by Dr. Roualdes, Dr. Donatello, and Dr. Gray.